HW#7

Homework Report

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Problem 01 7.3

We observe N IID samples from the PDFs:
a. Gaussian

$$p(x;\mu) = \frac{1}{\sqrt{2\pi}}\exp\left[-\frac{1}{2}(x-\mu)^2\right].$$

b. Exponential

$$p(x;\lambda) = \begin{cases}\lambda\exp(-\lambda x) & x > 0 \\ 0 & x < 0.\end{cases}$$

In each case find the MLE of the unknown parameter and be sure to verify that it indeed maximizes the likelihood function. Do the estimators make sense?

Solution:

(a) Gaussian distribution

Since the observations are IID, the likelihood function is

$$L(\mu;\mathbf{x}) = (2\pi)^{-N/2}\exp\left[-\frac{1}{2}\sum_{n=0}^{N-1}(x[n]-\mu)^2\right].$$

Taking the logarithm gives

$$\ell(\mu) = -\frac{N}{2}\ln(2\pi) - \frac{1}{2}\sum_{n=0}^{N-1}(x[n]-\mu)^2.$$

Differentiating with respect to μ,

$$\frac{d\ell}{d\mu} = \sum_{n=0}^{N-1}(x[n]-\mu) = \sum_{n=0}^{N-1}x[n] - N\mu.$$

Setting the derivative equal to zero yields

$$\hat{\mu}_{\mathrm{ML}} = \frac{1}{N}\sum_{n=0}^{N-1}x[n] = \bar{x}.$$

The second derivative is

$$\frac{d^2\ell}{d\mu^2} = -N < 0$$

Thus, the log-likelihood is strictly concave over the entire parameter space, and $\bar{x}$ is its unique global maximizer. Since the logarithm is strictly increasing, it also maximizes the likelihood.

The result is reasonable because $E(x[n]) = \mu$: the sample mean is a natural estimate of the population mean.

(b) Exponential distribution

For observations satisfying $x[n] > 0$, the likelihood function is

$$L(\lambda; \mathbf{x}) = \prod_{n=0}^{N-1} \lambda e^{-\lambda x[n]}$$
$$= \lambda^N \exp\left[-\lambda \sum_{n=0}^{N-1} x[n]\right], \lambda > 0.$$

The log-likelihood is

$$\ell(\lambda) = N \ln \lambda - \lambda \sum_{n=0}^{N-1} x[n].$$

Differentiating,

$$\frac{d\ell}{d\lambda} = \frac{N}{\lambda} - \sum_{n=0}^{N-1} x[n].$$

Setting this equal to zero gives

$$\hat{\lambda}_{\mathrm{ML}} = \frac{N}{\sum_{n=0}^{N-1} x[n]} = \frac{1}{\bar{x}}.$$

Because all observations are positive, this estimate satisfies $\hat{\lambda}_{\mathrm{ML}} > 0$. Furthermore,

$$\frac{d^2\ell}{d\lambda^2} = -\frac{N}{\lambda^2} < 0, \lambda > 0.$$

Hence, the log-likelihood is strictly concave, and the estimate is the unique global maximizer.
This result also makes sense because the exponential distribution has mean

$$E(x[n]) = \frac{1}{\lambda}.$$

Estimating $1/\lambda$ by the sample mean and taking its reciprocal gives $\hat{\lambda}_{\mathrm{ML}} = 1/\bar{x}$. Larger observed values therefore lead to a smaller estimated rate.

Problem 02 7.4
Asymptotic results can sometimes be misleading and so should be carefully applied. As an example, for two unbiased estimators of θ the variances are given by

$$\operatorname{var}(\hat{\theta}_1) = \frac{2}{N}$$
$$\operatorname{var}(\hat{\theta}_2) = \frac{1}{N} + \frac{100}{N^2}.$$

Plot the variances versus N to determine the better estimator.

Solution:

Both estimators are unbiased, so their mean squared errors equal their variances:

$$\operatorname{MSE}(\hat{\theta}_i) = \operatorname{Var}(\hat{\theta}_i).$$

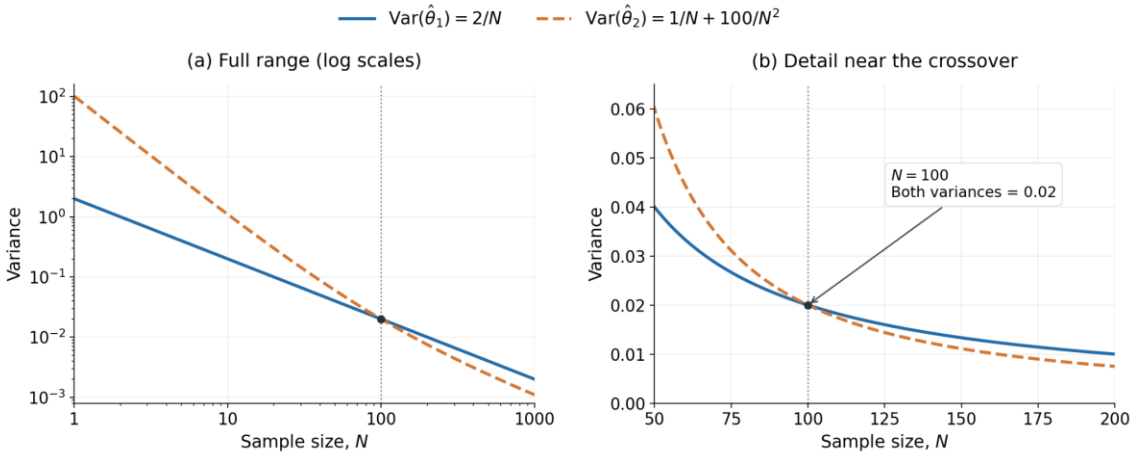
Thus, the estimator with the smaller variance has the smaller mean squared error. The difference between the variances is

$$\begin{aligned}\operatorname{Var}(\hat{\theta}_2) - \operatorname{Var}(\hat{\theta}_1) &= \frac{1}{N} + \frac{100}{N^2} - \frac{2}{N} \\ &= \frac{100 - N}{N^2}.\end{aligned}$$

Since $N^2 > 0$, the sign of this difference is determined by $100 - N$.

Sample size	Variance comparison	Better estimator
$1 \le N < 100$	$\operatorname{Var}(\hat{\theta}_1) < \operatorname{Var}(\hat{\theta}_2)$	$\hat{\theta}_1$
$N = 100$	Both variances equal 0.02	Equal performance
$N > 100$	$\operatorname{Var}(\hat{\theta}_2) < \operatorname{Var}(\hat{\theta}_1)$	$\hat{\theta}_2$

The variance curves are shown below. The left panel uses logarithmic axes to display the overall behavior; the right panel shows the crossover near $N = 100$.



As $N \to \infty$,

$$\mathrm{Var}\left(\hat{\theta}_1\right) = \frac{2}{N}, \mathrm{Var}\left(\hat{\theta}_2\right) \sim \frac{1}{N}.$$

Consequently,

$$\lim_{N\to\infty} \frac{\mathrm{Var}\left(\hat{\theta}_2\right)}{\mathrm{Var}\left(\hat{\theta}_1\right)} = \frac{1}{2}.$$

Although the second estimator is asymptotically better, it performs worse when $N < 100$. The importance of the higher-order term can be seen from

$$\frac{100/N^2}{1/N} = \frac{100}{N}.$$

Neglecting $100/N^2$ is justified only when N is sufficiently large relative to 100 . Therefore, superior asymptotic performance does not necessarily imply superior performance at a given finite sample size.

Problem 03 7.7
Consider N IID observations from the exponential family of PDFs

$$p(x;\theta) = \exp[A(\theta)B(x) + C(x) + D(\theta)]$$

where A, B, C, and D are functions of their respective arguments. Find an equation to be solved for the MLE. Now apply your results to the PDFs in Problem 7.3.

Solution:
For N IID observations, the likelihood function is

$$\begin{aligned} L(\theta;\mathbf{x}) &= \prod_{n=0}^{N-1} \exp[A(\theta)B(x[n]) + C(x[n]) + D(\theta)] \\ &= \exp\left[A(\theta)\sum_{n=0}^{N-1} B(x[n]) + \sum_{n=0}^{N-1} C(x[n]) + ND(\theta)\right]. \end{aligned}$$

Therefore, the log-likelihood is

$$\ell(\theta) = A(\theta)\sum_{n=0}^{N-1} B(x[n]) + \sum_{n=0}^{N-1} C(x[n]) + ND(\theta).$$

Assuming differentiability and parameter-independent support,

$$\frac{d\ell}{d\theta} = A'(\theta)\sum_{n=0}^{N-1} B(x[n]) + ND'(\theta).$$

An interior MLE must therefore satisfy the likelihood equation

$$A'\left(\hat{\theta}_{\mathrm{ML}}\right)\sum_{n=0}^{N-1} B(x[n]) + ND'\left(\hat{\theta}_{\mathrm{ML}}\right) = 0.$$

If $A'\left(\hat{\theta}_{\mathrm{ML}}\right) \neq 0$, this can also be written as

$$\frac{1}{N}\sum_{n=0}^{N-1} B(x[n]) = -\frac{D'\left(\hat{\theta}_{\mathrm{ML}}\right)}{A'\left(\hat{\theta}_{\mathrm{ML}}\right)}.$$

Any solution must be checked to ensure that it maximizes the likelihood; possible boundary maxima must also be considered.

Application to the Gaussian distribution

The Gaussian PDF can be expressed as

$$\begin{aligned} p(x;\mu) &= \frac{1}{\sqrt{2\pi}}\exp\left[-\frac{1}{2}(x-\mu)^2\right] \\ &= \exp\left[\mu x - \frac{x^2}{2} - \frac{\mu^2}{2} - \frac{1}{2}\ln(2\pi)\right]. \end{aligned}$$

Choose

$$\begin{gathered} A(\mu) = \mu, B(x) = x, \\ C(x) = -\frac{x^2}{2}, D(\mu) = -\frac{\mu^2}{2} - \frac{1}{2}\ln(2\pi). \end{gathered}$$

Then

$$A'(\mu) = 1, D'(\mu) = -\mu.$$

The likelihood equation becomes

$$\sum_{n=0}^{N-1} x[n] - N\hat{\mu}_{\mathrm{ML}} = 0$$

giving

$$\hat{\mu}_{\mathrm{ML}} = \frac{1}{N}\sum_{n=0}^{N-1} x[n].$$

Since $\ell''(\mu) = -N < 0$, this is the unique global maximizer.

Application to the exponential distribution
For $x > 0$,

$$p(x;\lambda) = \lambda e^{-\lambda x} = \exp[-\lambda x + \ln\lambda].$$

Choose

$$A(\lambda) = -\lambda, \quad B(x) = x,$$
$$C(x) = 0, \quad D(\lambda) = \ln \lambda.$$

Thus,

$$A'(\lambda) = -1, D'(\lambda) = \frac{1}{\lambda}.$$

The likelihood equation becomes

$$-\sum_{n=0}^{N-1} x[n] + \frac{N}{\hat{\lambda}_{\mathrm{ML}}} = 0,$$

so

$$\hat{\lambda}_{\mathrm{ML}} = \frac{N}{\sum_{n=0}^{N-1} x[n]} = \frac{1}{\bar{x}}.$$

Since $\ell''(\lambda) = -N/\lambda^2 < 0$ for $\lambda > 0$, this is also the unique global maximizer.

Problem 04 7.8
If we observe N IID samples from a Bernoulli experiment (coin toss) with the probabilities

$$\Pr\{x[n] = 1\} = p$$
$$\Pr\{x[n] = 0\} = 1 - p$$

find the MLE of p.

Solution:
Let

$$K = \sum_{n=0}^{N-1} x[n]$$

denote the total number of successes. By independence, the likelihood function is

$$L(p; \mathbf{x}) = \prod_{n=0}^{N-1} p^{x[n]}(1-p)^{1-x[n]}$$
$$= p^K (1-p)^{N-K}, 0 \le p \le 1.$$

For $0 < K < N$, the log-likelihood is

$$\ell(p) = K \ln p + (N-K)\ln(1-p).$$

Differentiating,

$$\frac{d\ell}{dp} = \frac{K}{p} - \frac{N-K}{1-p}.$$

Setting the derivative equal to zero gives

$$K(1-p) - (N-K)p = 0$$

or

$$K - Np = 0.$$

Therefore,

$$\hat{p}_{\mathrm{ML}} = \frac{K}{N}.$$

The second derivative is

$$\frac{d^2\ell}{dp^2} = -\frac{K}{p^2} - \frac{N-K}{(1-p)^2} < 0, 0 < p < 1.$$

Thus, this solution uniquely maximizes the likelihood when $0 < K < N$.
The boundary cases must also be considered:

- If $K = 0$, then $L(p) = (1-p)^N$, which is maximized at $p = 0$.
- If $K = N$, then $L(p) = p^N$, which is maximized at $p = 1$.

Hence, including the boundary cases, the MLE is

$$\hat{p}_{\mathrm{ML}} = \frac{K}{N} = \frac{1}{N}\sum_{n=0}^{N-1} x[n].$$

The estimated success probability is therefore the observed proportion of successes.